

Introduction to Prime Numbers

Alex Glandon

June 2026

1 Theorem 1

$A \neq B$, A and B both prime

$$\{1, 2, 3, \dots, B\} = \{A \% B, (2A) \% B, (3A) \% B, \dots, ((B-1)A) \% B\}$$

(as sets)

2 Theorem 2

$A \neq B$, A and B both prime

$$\forall C, \{1, 2, 3, \dots, B\} =$$

$$\{(A+C) \% B, (2A+C) \% B, (3A+C) \% B, \dots, ((B-1)A+C) \% B\}$$

(for Ms. Shannon)